

Heterogeneous graph-based knowledge tracing with spatiotemporal evolution

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ABSTRACT

Knowledge tracing (KT), in which the future performance of students is estimated by tracing their knowledge states based on their responses to exercises, is widely applied in the field of intelligent education. However, existing mainstream KT models explore the importance of knowledge relations but ignore the key role of cognitive factors. According to the knowledge construction theory, the human cognitive system performs both spatial accommodation and temporal assimilation to internalize knowledge. In this paper, we propose an innovative heterogeneous graph-based Knowledge tracing method with spatiotemporal evolution (TSKT), in which knowledge state evolution is traced along both temporal and spatial dimensions. We construct a heterogeneous graph with multiple exercise attributes, including content, concepts, and difficulty, to obtain a knowledge space with richer exercise representations through hierarchical aggregation. We design a spatial updating module in which each interaction updates the current node's state of the knowledge space and transfers its influence to its neighbors. We also design a temporal updating module to further update the knowledge state through short-term memory enhancing and long-term memory forgetting. Finally, we stack these modules to obtain deeper features by using alternate spatiotemporal updating. Extensive experiments on three datasets reveal the superiority of the proposed method and its variants in terms of future performance prediction.

1. Introduction

With the current continuous development in information and Internet technology, online education providers such as Khan Academy, Assistent, and Coursera have developed a large body of teaching resources that can overcome the time and space limitations faced by several students (Zhuang et al., 2022). Therefore, establishing a scientific measurement method to trace the knowledge status of students is a key issue for appropriate learning material suggestion and personalized learning path recommendation (Hooshyar et al., 2022) in intelligent tutoring systems, and this aspect has been attracting growing attention from the public (Gao et al., 2021; Huang et al., 2021; Liu et al., 2019).

Knowledge tracing (KT) (Liu et al., 2021) aims to use time-series data obtained from student answers to exercises in the learning process to model knowledge mastery and predict future performance, as illustrated

in Fig. 1(a). For example, deep knowledge tracing (DKT) (Piech et al., 2015) and dynamic key-value memory networks (DKVMN) (Zhang et al., 2017) use the hidden vectors of a recurrent neural network (RNN) and memory matrix, respectively, to trace the evolution of knowledge in the temporal dimension. Other studies have focused on the impact of forgetting factors on the knowledge state (Ghosh et al., 2020; Nagatani et al., 2019). In addition to the temporal dimension, an increasing number of KT methods (Nakagawa et al., 2019; Tong et al., 2020a, 2020b) have emphasized the importance of spatial dimension relationships between exercises and knowledge concepts. SKT (Tong et al., 2020a, 2020b) involved the use of an exercise-knowledge hierarchical graph neural network to describe these relationships. GKT (Nakagawa et al., 2019) required the construction of a homogeneous graph using prerequisite relations to represent the relation between knowledge concepts. JKT (Song et al., 2021) employed a joint graph convolutional

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network based on the implicit relationships of “exercise-to-exercise” and “concept-to-concept.” Although rich representations of exercises have significantly improved prediction performance, previous studies have ignored the spatiotemporal evolution of the knowledge states of students.

According to the knowledge construction theory adopted in the field of education (Kimmerle et al., 2010, 2015; Piaget, 1970, 1977), when students arrive at a new exercise, their cognitive system does not internalize knowledge by simply transferring and integrating the information (Piaget, 1970). Here, knowledge internalization is an active process of adding new content (assimilation) and changing the structure of the knowledge (accommodation) (Kimmerle et al., 2010). Assimilation and accommodation are natural cognitive laws that cause a student’s knowledge state to change over time and space. Assimilation occurs in the temporal dimension when students arrive at new exercises, ultimately adding new pieces of information to their prior knowledge (Kimmerle et al., 2010). For example, students answering a new exercise, e_2 , at time t_2 will experience decays in their knowledge state depending on the time interval, the degree of assimilation of their old knowledge (Piaget, 1977), and updating of their current knowledge state in the temporal dimension. Meanwhile, accommodation occurs in the spatial dimension when a student acquires knowledge and transfers it to other associated knowledge. For example, as shown in Fig. 1(c), when a student interacts with exercise e_1 , the knowledge states of related concepts k_2 and k_3 are updated, and the effect is transferred to similar knowledge via accommodation (Piaget, 1970). Consequently, according to the knowledge construction process that occurs with exercise answering, the modeling of KT must consider accommodation with transfer and assimilation with the reception of knowledge in both the temporal and spatial dimensions.

However, two main challenges are associated with this approach. First, the evolution of a student’s knowledge state follows the mechanism of assimilation and accommodation based on the knowledge construction theory. As depicted in Fig. 1(c), a student’s knowledge undergoes assimilation over time on the temporal axis, while on the spatial axis the knowledge status is accommodated by neighboring concepts within the knowledge structure. Specifically, for the spatial evolution, the state changes of each knowledge concept within the knowledge space are transferred to adjacent concepts. For instance, when a particular exercise e_1 is answered correctly (or incorrectly), the student’s proficiency of k_1 increases (or decreases). This effect will be transferred to its neighbors, that is, k_2 and k_3 , as well as to higher-level

associated knowledge concepts. Evidently, at time t_1 , the student’s mastery of knowledge concepts k_2 and k_3 have both diminished from their initial default value of 1. Moreover, for the temporal evolution, knowledge can only be assimilated and received if it is not forgotten over time. Considerably, at time t_2 , as time advances, the student’s proficiency in a specific knowledge concept (i.e., k_3 or k_4) has decreased, indicating forgetting without being assimilated or absorbed. However, forgetting is influenced by a variety of factors including the frequency of applying the same knowledge, and it is often difficult for a general RNN (Piech et al., 2015) to capture the influence of different time lengths on forgetting. Therefore, it is necessary to portray the assimilation and accommodation processes of knowledge in both the temporal and spatial dimensions.

Second, knowledge construction in temporal and spatial evolution is dependent on the relationship between exercises and knowledge concepts. The knowledge space theory (Doignon & Falmagne, 1985) asserts that an exercise is not isolated; instead, there exist relations between exercises resulting from prerequisite relations between knowledge (Chen et al., 2018). In fact, each exercise contains a variety of heterogeneous characteristics such as content q , knowledge concept c , and difficulty d . The exercise heterogeneous graph network comprises multiple heterogeneous nodes, as shown in Fig. 1(b). Consequently, a comprehensive exercise representation necessitates the fusion of diverse exercises and their heterogeneous attributes. Nonetheless, developing methodologies that effectively harness these heterogeneous attributes to create accurate exercise representations and capture their implicit knowledge relationships presents a significant challenge.

The primary aim of our research is to calculate students’ knowledge states by tracing the processes of assimilation and accommodation to address the inadequate knowledge construction in traditional models. Consequently, we divide the knowledge modeling process into two dimensions: temporal and spatial. Existing KT methods mainly focus on updating knowledge in the temporal dimension, lacking consideration of spatial influences that result in limited knowledge propagation. To address this challenge, we propose a novel heterogeneous graph-based KT method with temporal-spatial evolution (TSKT) to update and trace students’ knowledge states over time and space. As for the spatial dimension’s accommodation process, we extend the traditional homogeneous graph-based knowledge propagation approach (Tong et al., 2020a, 2020b) by employing graph neural networks to aggregate multi-dimensional heterogeneous information from exercises, thereby enriching the knowledge representation of exercises. Additionally,

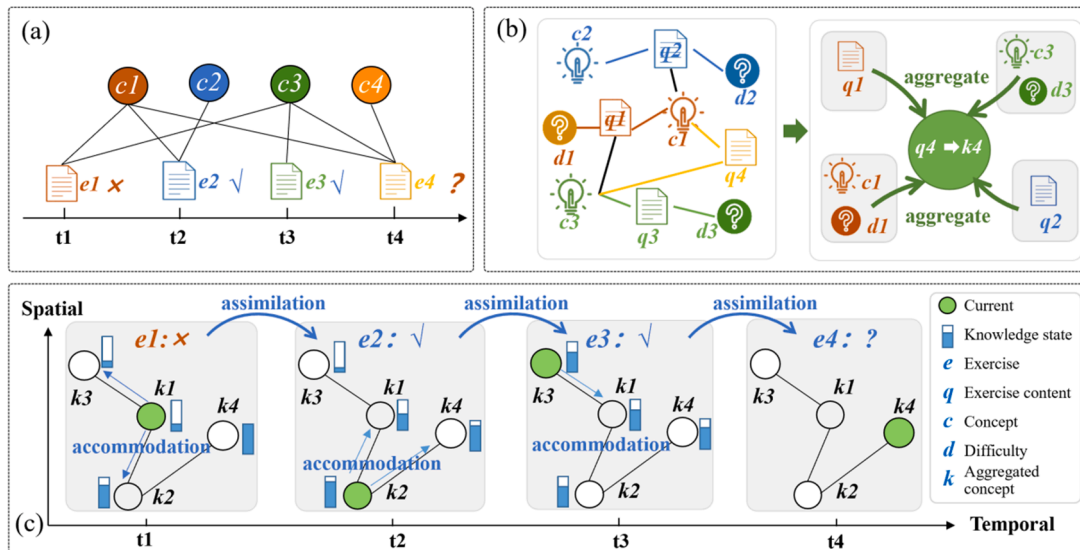


Fig. 1. A toy example. (a) The learning process in KT scenarios. (b) Multiple heterogeneous information of exercises. (c) Knowledge state evolution based on knowledge construction in both temporal and spatial dimensions. The default original knowledge status of each knowledge concept is 1.

beyond updating the current node state based on its current response, the influence is also propagated to neighboring nodes, inducing a response update among related nodes within the knowledge structure. As for the temporal dimension's assimilation process, we introduce three temporal sequence features and leverage forgetting rules to compute changes in latent knowledge. We summarize the contributions of our approach as follows:

- (1) We first construct a heterogeneous graph network with exercise content, knowledge, and difficulty, utilizing a hierarchical aggregation method to acquire representations for the diverse exercise nodes.
- (2) For the spatial dimension, we establish a spatial updating module to update students' current knowledge states and transfer the accommodation influence of relevant knowledge through neighboring relations in the knowledge space.
- (3) For the temporal dimension, we develop a gate-based updating mechanism to simulate forgetting, incorporating temporal factors like sequence time intervals, repeated time intervals, and past trial counts to obtain assimilated knowledge.

In our analysis, we conduct extensive experiments on three datasets to evaluate the performance of TSKT, comparing it with existing KT models. We also perform ablation experiments to clarify the importance of each key component. Finally, we employ clustering and evolution visualization to illustrate the rationality and interpretability.

2. Related work

This section provides a brief review of relevant literature based on the following three aspects: knowledge tracing, exercise representation, and graph neural networks.

2.1. Knowledge tracing

Currently, several deep learning methods have been applied to KT. In particular, DKT (Piech et al., 2015) first introduced deep learning into KT using hidden vectors in RNNs to represent the hidden student knowledge mastery state. Following this, the DKVMN (Zhang et al., 2017) extended the knowledge hidden vector of DKT using a fixed-size memory matrix to store the knowledge state at each moment to capture the relationship between knowledge concepts and exercises. Notably, numerous variations of the DKVMN have been developed. Sequential key-value memory networks (Abdelrahman & Wang, 2019) replace the main component of the DKVMN with an improved version of Hop-LSTM. DKVMN-CA (Ai et al., 2019) introduces mapping relations between exercises and knowledge concepts into the DKVMN model. Therefore, DKVMN-based models can automatically learn the relations between exercises and multiple latent knowledge concepts, encouraging researchers to focus on the knowledge structure. To evaluate the influence of the knowledge structure on KT tasks, prerequisite-driven deep knowledge tracing (Chen et al., 2018) incorporates question-concept mapping, student-question matrices, and prerequisite relations between knowledge concepts into DKT. To better represent knowledge relationships, more and more researchers are focusing on graph KT models, such as GKT (Nakagawa et al., 2019), GIKT (Yang et al., 2021), and SGKT (Sun et al., 2022). They are using a graph convolutional network (GCN) to aggregate the information between exercises and concepts, to realize knowledge tracing through an RNN. In fact, it is difficult for current graph KT to account for both the knowledge concept relations hidden in the multi-source heterogeneous features of exercises and the spatiotemporal evolution of knowledge states, thus limiting their performance to some extent.

2.2. Exercise representation

Exercises are tools used to quantify a student's knowledge mastery level in education assessment. A reasonable exercise representation is expected to effectively promote the prediction performance of KT. DKT (Piech et al., 2015) uses knowledge concepts instead of exercises. The DKVMN (Zhang et al., 2017) and dynamic knowledge diagnosis approaches integrating cognitive features (Huang et al., 2021) use the combination method of exercise embedding and key-value memory network to represent an exercise as high-dimensional vector. The previously described representation methods are simply based on knowledge concepts, resulting in substantial loss of information. An increasing number of researchers are focusing on the significance of content. For example, Huang et al. (2020) proposed that a richer exercise representation is useful for fine-grained and multi-label text classification. In the first attempt to utilize the semantic information of content to represent exercises, an exercise-enhanced RNN (EERNN) (Liu et al., 2019) was applied by encoding each word into a word embedding vector via word2vec. QuesNet (Yin et al., 2019), based on the EERNN, is a unified framework used to aggregate exercise information with heterogeneous information from different attributes of exercises. A graph-based interaction model for KT (GIKT) (Yang et al., 2021) aggregates the information of exercises and knowledge concepts through GCN, pioneering the association-driven knowledge tracing as illustrated in Fig. 1(b). Inspired by GIKT, APGKT (Zhang et al., 2022) constructs a skill associative path from exercises graph and skills graph to capture a higher dimensional representation of exercises. Ensemble KT (EnKT) (Sun et al., 2022) is a hybrid of Bayesian knowledge tracing and DKT. It models the relationships among exercises, concepts, and students. Context-aware knowledge tracing integrated with exercise representation (ERAKT) traces knowledge state by combining a mathematical representation of exercises with exercise association (Huang et al., 2021), incorporating formula information and the BERT model (Devlin et al., 2019) to obtain text embedding representations of exercises.

2.3. Graph neural networks

Graph neural networks can be classified into homogeneous and heterogeneous graphs, based on the types of nodes and edges. In particular, homogeneous graphs have only one type of node and one type of relation. Graph convolutional networks (Kipf & Welling, 2017) are the most basic graph neural networks. In these networks, the convolutional operator is introduced to allow aggregation of the features of neighboring nodes into the representation of individual nodes, thereby achieving a degree of convergence of the information of all nodes. The graph sample and aggregate (GraphSAGE) (Ahmed et al., 2017) approach employs the LSTM architecture for information aggregation and node-centric neighbor sampling. A graph attention network (GAT) (Velićković et al., 2018) is based on the self-attention mechanism of Transformer (Vaswani et al., 2017), which allows different weights to be assigned to nearby nodes depending on their significance. In the domain of knowledge tracing, GKT (Nakagawa et al., 2019) involves the construction of homogeneous graphs based on the prerequisites of the knowledge concepts associated with the exercises, which enhance the knowledge representation. When a student completes an exercise, the knowledge concept nodes and nearby nodes in the corresponding graph structure are updated at each time step. Heterogeneous graphs have a variety of node and relation types. Because real-world node-edge interactions are generally heterogeneous and contain more information, heterogeneous graphs are more widely employed than homogeneous graphs (Li et al., 2017). Session GKT (SGKT) (Sun et al., 2022) uses gated graph neural networks to obtain the students' knowledge state from the relationship graph session between exercises and skills. In the academic field, Zhang introduced a heterogeneous graph neural network (Hong et al., 2020; Zhang et al., 2019) model based on the different relations between three heterogeneous nodes: paper, author, and avenue. Liu

et al. (2022) proposed the linear graph latent ability model utilizing graph-structured interactions to construct student and item latent ability factors. Therefore, heterogeneous graphs can aggregate more types of feature information than homogeneous graphs, resulting in a more comprehensive representation of exercises.

3. Problem formulation

In this section, we introduce the primary concepts involved in KT, such as the definition of the knowledge space, and introduce the task and definition of KT based on the knowledge space.

3.1. Knowledge space

According to the educational measurement theory, exercises have a variety of attributes including difficulty (Wang et al., 2020), knowledge concept (Tong et al., 2020a, 2020b), and exercise content (Huang et al., 2021; Liu et al., 2019). These multi-source heterogeneous features enrich the representation of exercises (Yin et al., 2019). However, the student's knowledge state is stable over time, and the responses provided by students to related exercises are consistent. Consequently, the knowledge space is a homogeneous graph because it only contains nodes of the same type and their relations, whereas the exercise heterogeneous graph, as shown in Fig. 1(b), can aggregate a wide range of exercise attributes.

Definition 1. (Exercise heterogeneous graph) Given a heterogeneous graph, $G = (V, C, O_V, R_C)$, where V represents multiple types of nodes, C represents the edges between different nodes, O_V is the node type set, and R_C is the edge relation set.

Definition 2. (Knowledge space) Given a knowledge space, nodes represent individual exercises, and edges are the relations between nodes. Thus, we define the knowledge space as graph $S = (E, C)$, where

$E = \{e_1, e_2, \dots, e_N\}$, and each node represents a different exercise, $e_i, i = 1, \dots, N$, where N is the number of exercises, and C is the edge between nodes.

3.2. Problem statement

We apply the relations in knowledge space S to a KT task by modeling the prior knowledge state of students based on their responses to different exercises. The influence of knowledge relation and future performance must be explored.

Definition 3. (KT with knowledge space) Given an interactive sequence of student interactions over a certain time period, the response interaction cell, $x_t = (e_t, r_t)$, denoting a student completing exercise e_t ($e_t \in \{e_1, \dots, e_N\}$) at time t , which is the vertex in knowledge space S , and the corresponding response is r_t , which is a binary variable, $r_t \in \{0, 1\}$. When a student answers correctly, $r_t = 1$; otherwise, $r_t = 0$. Our goal is two-fold: (1) model the student's knowledge state for all exercises (i.e., exercise nodes E in S) and (2) predict the probability that the student provides correct responses to a new exercise e_{t+1} in the future, which is represented by $P(e_{t+1} = 1 | e_{t+1}, X, S)$.

4. TSKT method

This section begins with our new solution overview of TSKT, and following this, the components of TSKT are introduced in detail.

4.1. Overview

In this section, we provide a comprehensive overview of the TSKT model, which is developed based on the assimilation and accommodation mechanism of knowledge states depicted in Fig. 1(c). An illustrative depiction of TSKT is presented in Fig. 2; it comprises three modules:

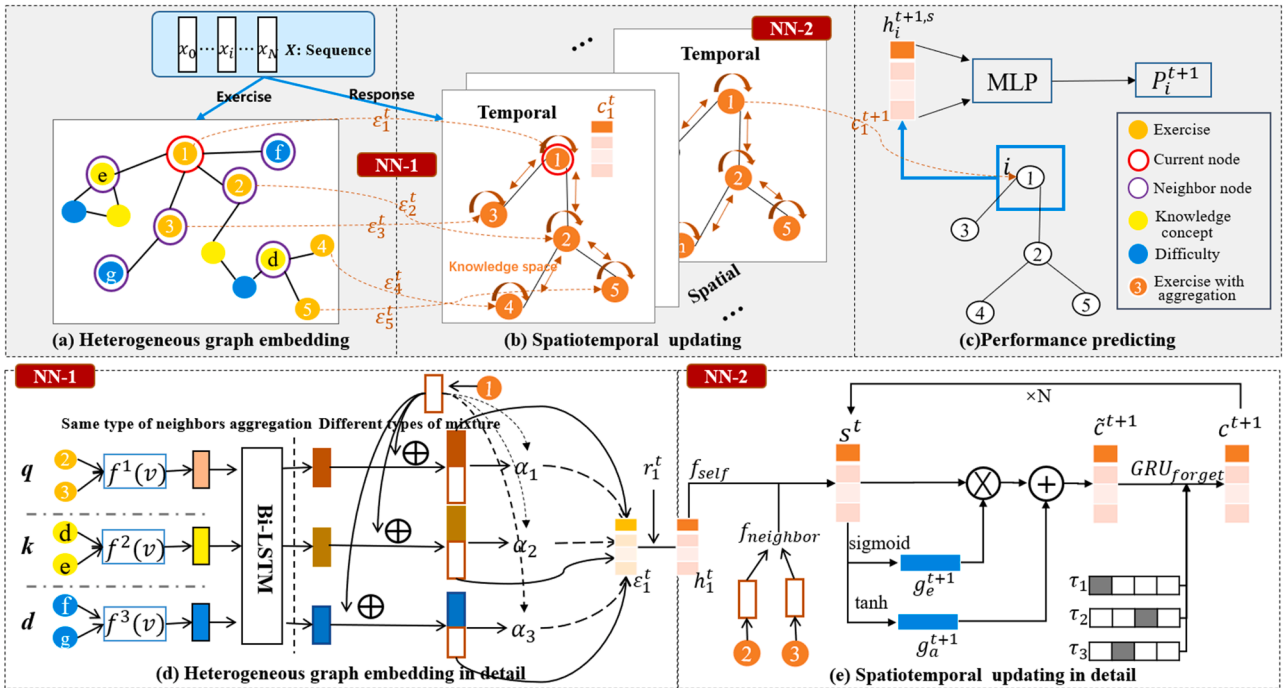


Fig. 2. Overview of TSKT framework. TSKT is generally composed of three modules: (a) heterogeneous graph embedding, (b) spatiotemporal updating, and (c) performance predicting. The detailed process for the first two modules is named NN-1 and NN2, shown in (d) and (e). For the first module, an exercise heterogeneous graph is constructed to aggregate different information on exercise content, knowledge concept, and difficulty, outputting the exercise representation e_t^i . Then, the students' knowledge state h_t^i can be captured by adding the response r_t^i , which is updated in both temporal and spatial dimensions in the knowledge space by applying alternate spatiotemporal updating. Thus, we obtain the updated potential knowledge state of students, c_i^{t+1} . Finally, we feed c_i^{t+1} into an MLP algorithm to predict the future performance.

heterogeneous graph embedding (as shown in Fig. 2(a)), spatiotemporal updating (as shown in Fig. 2(b)), and performance predicting modules (as shown in Fig. 2(c)). Given that exercises contain multiple heterogeneous information attributes such as difficulty, content, and knowledge concepts, we construct heterogeneous graph G to obtain the exercise representation through the same type of neighbor aggregation and different types of mixture, as shown in Fig. 2(d). To simulate the evolution of a student's knowledge state with exercises in both the temporal and spatial dimensions, we propose a temporal update component based on the forgetting curve and a spatial update component based on the aggregation of information, considering the relation between knowledge and exercise, shown in Fig. 2(e). Finally, according to the student's current knowledge state, a multilayer perceptron (MLP) is used to predict whether the student will be able to answer a future exercise correctly. The procedure is explained in detail as follows:

4.2. Heterogeneous graph embedding for exercises

A heterogeneous graph based on Definition 1 can be defined as $G = (V, C, O_V, R_C)$, where O_V represents different types of nodes, such as knowledge concept, exercise content and exercise difficulty. We obtain the exercise embedding matrix $\varepsilon \in \mathbb{R}^{|V| \times d}$ through the heterogeneous graph. The steps are as follows:

- 1) *Encoding of node information.* We apply two different encoding methods. We adopt one-hot encoding for structured data such as knowledge concepts and exercise difficulty because they are categorical. For unstructured data such as text content of the exercise, which contains words and formulas, a pretrained BERT model (Devlin et al., 2019) is used to encode text features.
- 2) *Meta-path sampling.* We start from node V in the heterogeneous graph and randomly select neighbor nodes of the current node as the next walking target according to a certain probability. This cycle is repeated until the walking path reaches a pre-set length. We then divide the nodes along the path into three groups according to their types, count them, and select a certain number of nodes with the highest frequency as the adjacent nodes of the current node.
- 3) *Aggregation.* To aggregate the heterogeneous neighbor information of each node, we adopt the hierarchical aggregation method. The details are shown in Fig. 2(d). We first aggregate the node information of the same type separately, and then aggregate the node information of different types through the attention mechanism.

Aggregation of same type. To effectively aggregate different types of node information, it is essential to classify and process nodes by aggregating information of the same type initially. However, direct concatenation or summation of node features (Hamilton et al., 2017; Velićković et al., 2018) can lead to information loss. Drawing inspiration from prior work on aggregating unordered neighbors (Hamilton et al., 2017; Zhang et al., 2019), we employ the feature extraction capabilities of Bi-LSTM to compute type vectors. Specifically, we set up the Bi-LSTM to aggregate information for nodes of the same type and then obtain type vectors through average pooling. The formula is as follows:

$$f^i(v) = \sum_{w \in n^i(v)} \frac{[\overrightarrow{LSTM}(emb(w)), \overleftarrow{LSTM}(emb(w))]}{|n^i(v)|} \quad (1)$$

where $n^i(v)$ is the set of neighbor nodes of type i for the current node, $emb(w)$ represents an embedding representation of neighbor node w , and $f^i(v)$ is the aggregated vector of neighbor nodes of type i for node v . $\overrightarrow{LSTM}(\cdot)$ is the forward hidden sequence, and $\overleftarrow{LSTM}(\cdot)$ is the backward hidden sequence. $|\cdot|$ represents the number of the set.

Aggregation of different types. For different types of neighbor nodes, we adopt an attention mechanism to assign different weights to each aggregation vector and then combine them with the embedded

representation of node v . The formula v for weight calculation is as follows:

$$f_a(v_1, v_2) = \exp\{ReLU(\vartheta \cdot [v_1 \oplus v_2])\} \quad (2)$$

$$\alpha_{(v,i)} = \frac{f_a(emb(v), f^i(v))}{\sum_{w \in \{emb(v) \cup f^i(v), i \in O_V\}} f_a(emb(v), w)} \quad (3)$$

where ϑ represents trainable parameters, and $f_a(\cdot)$ is a function for calculating the correlation weight and $\alpha_{(v,i)}$ refer to the weight between two vectors. Then, we use the weights to aggregate the type vectors. The formula is as follows:

$$\varepsilon_v = \alpha_{(v,v)} \cdot emb(v) + \sum_{i \in O_V} \alpha_{(v,i)} \cdot f^i(v) \quad (4)$$

where ε_v refers to the aggregated vector representation of node v , and we use the exercise embedding matrix ε for the initial embedding and updating of nodes in the knowledge space.

4.3. Spatial updating

At each time step, the student answers the exercise and generates a corresponding response. We set vectors to record the student's mastery on each exercise, and each interaction updates the corresponding exercise node. Moreover, in the spatial dimension, the relation between knowledge concepts is expected to lead to the accommodation of knowledge states. The interaction at each time step not only affects the knowledge concepts associated with exercises but also transfers the influence to neighboring knowledge concepts. For example, as shown in Fig. 1(c), the effect of knowledge concept k_1 will be transferred to its neighbors, that is, k_2 and k_3 .

4.3.1. Construction of the knowledge space

We first construct an exercise homogeneous graph through the correlation between exercises, that is, knowledge space S . One node in the knowledge space corresponds to one exercise, indicating the student's mastery of the exercise. We use the exercise heterogeneous graph embedding ε to initialize the node embedding. Considering that the student's knowledge state is closely related to the knowledge concepts associated with the exercise, we set up edge connections between exercises with the same knowledge concepts. We use adjacency matrix $A \in \mathbb{R}^{N \times N}$ to represent the edge connections in the knowledge space, that is, A_{ij} represents the relationship between exercise i and exercise j . Specifically, A is normalized so that its values are all in interval $[0, 1]$, where 0 implies no edge, and non-zero values indicate the presence of an edge. The knowledge space is constructed based on the relationships between the knowledge concepts associated with the exercises and the nodes in the space representing the mastery of each exercise.

4.3.2. Current node update

At each time step, an interaction occurs in the knowledge space. Interactive information updates the corresponding answering exercise node e^t and neighbor nodes. Updating the exercise node involves two elements: the initial and interactive representation of the node. We combine the two and set up an MLP to complete the update. The formula is as follows:

$$h_e^t = [h_e^{t-1}, (r^t + e^t)] \quad (5)$$

$$s^t = f_{self}(h_e^t) \quad (6)$$

where h_e^{t-1} is the hidden state of the current node at the previous time step in the knowledge space, h_e^t is the intermediate vector to be updated, $r^t \in \mathbb{R}^d$ refers to answering the exercise correctly or incorrectly at time t , $e^t \in \mathbb{R}^d$ is the embedded representation of the corresponding answering

exercise at time t , f_{self} is an MLP, and s^t is the potential representation after updating.

4.3.3. Neighbor node update

At time t , the student answers the exercise; following this, accommodation occurs on all the neighbors of node e^t , denoted as $n(e^t)$. Because accommodation is the process of spreading the knowledge state, it is necessary to simulate the transfer of the knowledge state through the hidden knowledge state of the current node and the hidden representation of the adjacent node. We first splice the hidden knowledge state of e^t with the exercise of the neighbor node $j \in n(e^t)$, set up a new MLP to process the spliced vector, and use the value of the corresponding adjacency matrix as the weight. The updated specific formula is as follows:

$$h'_j = [[h_j^{t-1}, e^{t-1}], [h_e^{t-1}, (r^t + e^t)]], j \in n(e^t) \quad (7)$$

$$s'_j = A_{e^t, j} \times f_{neighbor}(h'_j) \quad (8)$$

where h_j^{t-1} refers to the hidden state of neighbor node j at the previous time step, $t-1$, h'_j is the intermediate vector to be updated, and $A_{e^t, j}$ refers to the weight value of the edge between current node e^t and neighbor node j ; $f_{neighbor}$ is an MLP with new parameters, and s'_j is the hidden vector of neighbor node j after updating.

4.4. Temporal updating

The cognitive system simulates the assimilation of knowledge by a student according to the interactions involved in answering exercises (Kimmerle et al., 2010) to update the corresponding knowledge state. Therefore, after spatial updating, the temporal updating layer assimilates and further updates the knowledge state through short-term memory updating and long-term memory forgetting. For example, as shown in Fig. 1(c), at time t_2 , due to the forgetting effect, the student's proficiency in a specific knowledge concept (i.e., k_3 or k_4) has decreased. Furthermore, we also stack spatial and temporal updating layers to obtain deeper features through alternate spatiotemporal updates.

4.4.1. Short-term memory updating

Inspired by the gate mechanism involved in the LSTM and DKVMN (Zhang et al., 2017), we consider the latent knowledge state vector of the exercise as input, set the erasure gate g_e , and the update gate g_a to control the transmission of information through the neural network. We use the weight vector obtained from the gate to update the knowledge state, which simulates assimilation of the student's filtering and enhancement of the current knowledge in the short term. The short-term update method is described in the following formula:

$$g_e^{t+1} = \text{Sigmoid}(W_e s^{t+1} + b_e) \quad (9)$$

$$g_a^{t+1} = \tanh(W_a s^{t+1} + b_a) \quad (10)$$

$$\tilde{c}^{t+1} = (1 - g_e^{t+1})s^{t+1} + g_a^{t+1} \quad (11)$$

where s^{t+1} refers to the latent knowledge state vector corresponding to exercise e^{t+1} at time $t+1$, W_e and W_a are the weights matrices, b_e and b_a are the bias matrices, and $\tilde{c}^{t+1}$ is the potential knowledge state after short-term memory update, which is a temporary intermediate vector.

4.4.2. Long-term memory forgetting

According to the knowledge construction theory, knowledge that has not been forgotten can be assimilated by new knowledge. Ebinho's forgetting theorem (Fogliatto & Anzanello, 2016; Huang et al., 2020) states that as the interval time becomes longer, the students' mastery of knowledge attenuates. According to the cognitive processing theory

(Pelánek, 2015), students strengthen the knowledge they have mastered through repeated practice to make the knowledge less easy to forget. Therefore, at time t , we consider three forgetting factors (Huang et al., 2021; Nagatani et al., 2019): (1) a sequence time interval τ_1^t , that is, the time interval of adjacent interactions in the answering sequence; (2) interval between repetitions τ_2^t , that is, the interval time between the last practice on the same knowledge concept and current interaction; and (3) past trial counts τ_3^t , that is, the time for which the knowledge concept has been practiced in the answering sequence.

We use the splicing method to combine the above three forgetting features with the knowledge state variable $\tilde{c}^{t+1}$ after the short-term memory update. To simulate the assimilation of memory and the conversion of short-term memory to long-term memory, we use a gated recurrent unit (GRU) (Chung et al., 2014) to fuse the forgetting information and further update the potential knowledge state. In the time dimension, the hidden knowledge state after assimilation is as follows:

$$c^{t+1} = \text{GRU}_{\text{forget}}(\tilde{c}^{t+1}, \tau_1^t, \tau_2^t, \tau_3^t) \quad (12)$$

4.4.3. Spatiotemporal evolution

The evolution of knowledge states occurs in the spatiotemporal dimension, as shown in Fig. 1(c). The update of the hidden vector of the knowledge state is carried out through the edge of the exercise node (Nakagawa et al., 2019). Therefore, we increase the number of temporal and spatial update rounds to better simulate the evolution. In fact, the knowledge state will be updated alternately from the spatial and temporal dimensions to aggregate the information of the more distant exercise nodes to obtain local, more complex, and higher-level graph features. Its update process is shown in Fig. 3.

In each cycle, the knowledge state vector of the node repeatedly undergoes the accommodation update in the spatial dimension and assimilation update in the temporal dimension. Consequently, the forgetting information and the spatial structure information of the exercises are continuously aggregated. The input of the temporal update layer of each cycle is the output of the spatial update layer of the previous cycle. The pseudo code is shown in Algorithm 1.

Algorithm 1: Spatiotemporal evolution.

Input: Knowledge Space $S = (E, C)$, updating layer L , exercise answering sequence $x^t = (e^t, r^t)$, adjacency matrix connection A .

Output: $\{c^{t,L} | t = 1, 2, \dots, T-1\}$

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1 for  $t = 1$  to  $T-1$  do
2   for  $l = 1$  to  $L$  do
3     if the node  $e$  is current interaction at  $t$ 
4       then
5         Update  $s^{t,l}$  using  $h_e^{t-1,l-1}$  via
6           Eq. (5) and (6);
7       else
8         for  $j$  in neighbor node  $n(e^t)$  do
9           Update  $s_j^{t,l}$  using  $h_e^{t-1,l-1}$ ,
10              $h_j^{t-1,l-1}$  and  $A_{e^t,j}$  via Eq. (7)
11             and (8);
12         end
13       end
14     end
15   end
16   Compute  $g_e^t$  and  $g_a^t$  via Eq. (9) and (10);
17   Update  $c^{t,L}$  using  $s^{t,L}$  via Eq. (11) and (12);
18 end

```

4.5. Future performance prediction

In the final prediction, the model outputs the future performance of the student in answering new exercises. We use the hidden state c^{t+1} of the corresponding exercise node as input and output the probability of

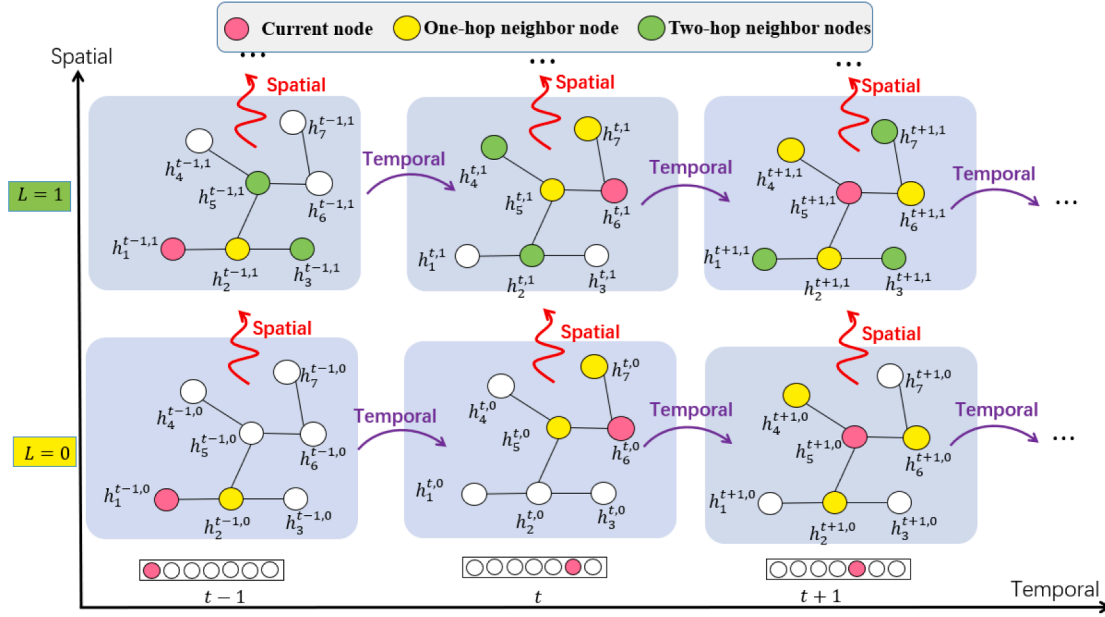


Fig. 3. Graph network updating in the spatiotemporal dimension: L is number of spatial cycles, and t is the number of temporal cycles. $h_i^{t+1,l}$ is the hidden state of node i at step t after the propagation of the neighbor influence in layer l . The knowledge state is updated from left to right in the temporal dimension and from bottom to top in the spatial dimension.

Table 1
Statistical Properties of Datasets.

Dataset	Assistments2009	Assistments2015	Eanalyst
Interactions	224,218	708,631	409,161
Students	2,921	19,917	1,652
Exercises/knowledge concepts	16,892/110	-/100	2,691/782
Interactions per student	77	59	248

answering correctly, that is, $p^{t+1} \in [0,1]$. The higher the value of p^{t+1} , the higher is the probability that the student will answer the exercise correctly. The method of prediction is shown in the following formula:

$$p^{t+1} = \text{Sigmoid}(W_{\text{pre}} c_i^{t+1,L} + b_{\text{pre}}) \quad (13)$$

To optimize our model, we use gradient descent to update the model parameters, minimizing the cross-entropy loss function between the predicted and the true answer. The objective function is defined as follows:

$$L = - \sum_{i \in \mathcal{Y}} (r^{t+1} \ln p^{t+1} + (1 - r^{t+1}) \ln (1 - p^{t+1})) \quad (14)$$

where r^{t+1} is the response of the current student to the exercise at time $t + 1$.

5. Experiments

5.1. Datasets

To evaluate the performance of our TSKT method, we conducted experiments using three datasets. Given that this model requires the integration of three distinct feature categories, they encompass exercise features (encompassing exercise content, knowledge concepts and difficulty), student-exercise interaction features, and forgetting factors (including sequence time interval, interval between repetitions, and past trial counts). The corresponding statistical details provided in Table 1, and the distribution of features within the dataset is displayed in Table 2.

Assistment2009¹ is derived from Assistments,² an online teaching assistant platform that gathered data on students' online mathematics learning and evaluation in 2009. Note that some model results on this dataset are missing, as the temporal information of the interaction is not provided.

Assistment2015³ is derived from mathematics assessment data gathered on the Assistments platform in 2015. On average, this database has the most students and the fewest interactions per student. In this dataset, the knowledge concept index is employed instead of the exercise index because the number of interactive exercises is not supplied.

Eanalyst is delivered by a platform that gathered assessment interaction data from primary and secondary school students in math in 2018. This dataset has comprehensive features, covering all the features in Table 2. Fig. 4(b) shows the distribution of the number of knowledge concepts for exam exercises. Approximately 15% of the exercises are related to multiple knowledge concepts, with an average of 1.3 knowledge concepts relating to each exercise. This dataset has the largest average number of student interactions, as shown in Fig. 4(a).

5.2. Experimental setup

In this section, we clarify the implementation details of our TSKT

¹ <https://sites.google.com/site/assistmentsdata/home/assistent-2009-2010-data>.

² <https://new.assistments.org/>.

³ <https://sites.google.com/site/assistmentsdata/home/2015-assistments-knowledge-builder-data>.

Table 2
Characteristics of every datasets.

Dataset	Exercises			Interactions	Forgetting factors		
	Content	Concept	Difficulty		Sequence time interval	Interval between repetitions	Past trial counts
Assistments2009	–	✓	✓	✓	–	–	–
Assistments2015	–	✓	✓	✓	✓	✓	✓
Eanalyst	✓	✓	✓	✓	✓	✓	✓

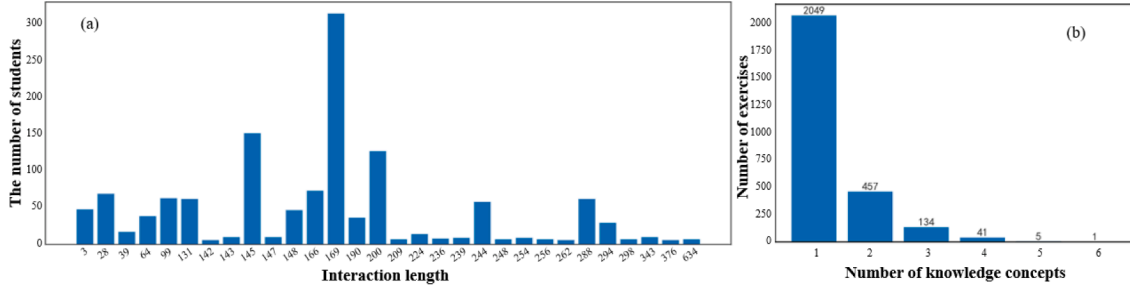


Fig. 4. (a) Interaction sequence length distribution in Eanalyst. (b) Distribution of knowledge concepts in Eanalyst.

method and introduce comparative baselines and evaluation metrics.

1) Implementation details. We used Pytorch-1.7.1 and cuda-toolkit-10.1.0 as our base development environment. The hardware configuration was a CPU Intel core i9-10920X and GPU GeForce RTX 3090. We used BERT to extract text features from exercises and set each exercise to a dimension of 128 in our heterogeneous graph. To alleviate overfitting, we applied dropout with $keep_prob = 0.1$. We set the learning rate as $1e-3$ and batch size as eight. In terms of the datasets, the training, validation, and test sets were split at a ratio of 7:2:1.

2) Comparison baselines. We compared our TSKT with several previous outstanding methods as our baselines:

DKT (Piech et al., 2015): DKT applies an RNN to model the learning process of every student, which can extract the representation of different student knowledge states.

DKVMN (Zhang et al., 2017): The DKVMN introduces the relationship between exercises and latent knowledge concepts to model student's knowledge states with a key-value matrix to store and update the states.

DKT + F (Nagatani et al., 2019): DKT + F is an improved variant of DKT, which introduces the forgetting behavior of students including intervals and repetition counts.

DKVMN + F: DKVMN + F is an improved variant of the DKVMN, which introduces the forgetting behavior of students and includes intervals and repetition counts.

GKT (Nakagawa et al., 2019): GKT maps knowledge concepts into nodes in the graph structure, aggregates each concept with its neighbors, and then carries out the classification task to model the student mastery level of each concept.

Our proposed model can construct a knowledge space for capturing the rich information of exercises by employing the heterogeneous graph neural network, which realizes the knowledge assimilation and accommodation by utilizing the alternate spatiotemporal updating method. TSKT depends on exercises with multi-dimensional heterogeneous attributes and three forgetting features. Therefore, to comprehensively verify the effectiveness of our TSKT model, we use the method of adding key modules one by one to design the variants. In the absence of features, the variants can replace our model to compare the performance, which enhances the generalization of the model. The variants of our TSKT model are as follows:

FKT: FKT uses three forgetting features in our TSKT model and updates the state of hidden knowledge by employing short-term and long-term memory components based on the forgetting curve theory in the

knowledge graph. Compared with TSKT, FKT can work when the heterogeneous exercise attributes are not available, demonstrating the effective of the assimilation in the temporal dimension.

SKT: SKT utilizes our spatial updating module in our TSKT model on the knowledge structure graph. It updates the knowledge status of the current node and neighboring nodes. Compared with TSKT, SKT can work without the forgetting features and the heterogeneous exercise attributes. It can further demonstrate the impact of the accommodation in the spatial dimension.

FSKT: FSKT introduces a temporal updating module based on forgetting, a spatial updating module according to the aggregated neighbor information into knowledge graph, and as well as an alternate spatiotemporal updating component in our TSKT model. Compared with TSKT, FSKT can work in the absence of the heterogeneous exercise attributes, which validate the effective of the spatiotemporal updating component.

3) Evaluation metrics. We used the area under curve (AUC) and accuracy (ACC) as our evaluation metrics, which are widely applied in KT problems. The values of AUC and ACC are between 0 and 1. High values indicate good results.

5.3. Experimental results

5.3.1. Performance prediction

To evaluate the prediction performance of our TSKT method, we compared it with five baseline models and three variants. The results on three datasets in terms of two evaluation metrics are shown in Table 3. We note some interesting observations. First, our TSKT method and its variants achieved the best performance both in terms of the AUC and ACC on all datasets. The variant SKT was 0.83% higher than the second-best model in terms of the AUC on the Assistment2009 dataset, and the other variant FSKT achieved the best performance in terms of both metrics on the Assistment2015 dataset. Our TSKT method outperformed all other variants and baselines on the Eanalyst dataset, which indicates that TSKT could better extract the relations between knowledge and sequential information. Second, the DKVMN (Zhang et al., 2017) performed better than DKT, which further verifies that considering the relation between exercise and potential knowledge concepts helps to improve the accuracy of prediction tasks. However, GKT demonstrated better prediction ability than the DKVMN by considering the hidden relations between knowledge concepts in the exercises to provide richer information for the update of students' knowledge status. Furthermore,

Table 3

Performance comparison on the KT task with baselines and variants.

	Methods	Forgetting	Spatial	Assistment2009		Assistment2015		Eanalyst	
				AUC	ACC	AUC	ACC	AUC	ACC
Baselines	DKT	–	–	0.7463	0.6689	0.6904	0.7275	0.8132	0.7565
	DKVMN	–	–	0.7748	0.7459	0.6969	0.7205	0.8283	0.7808
	DKT + F	✓	–	–	–	0.7027	0.7291	0.8197	0.7599
	DKVMN + F	✓	–	–	–	0.7036	0.7246	0.8542	0.7978
	GKT	–	–	0.7731	0.7360	0.6901	0.7269	0.8307	0.7800
Variants	FKT	✓	–	–	–	0.6990	0.7278	0.8427	0.7827
	SKT	–	✓	0.7814	0.7440	0.6988	0.7285	0.8514	0.7935
	FSKT	✓	✓	–	–	0.7053	0.7297	0.8500	0.7925
	TSKT	✓	✓	–	–	–	–	0.8617	0.8001

the variant SKT achieved better predictions than GKT (Nakagawa et al., 2019) by aggregating further-neighbor information in higher-level graph relations, which enhanced the accommodation process. Finally, it is also apparent that the model considering the time factor (DKT + F (Nagatani et al., 2019), DKVMN + F, and FKT) significantly outperformed the original model (DKT, DKVMN, and GKT) (Nakagawa et al., 2019) in terms of the AUC and ACC metrics, indicating the effectiveness of tracing the knowledge state decay process over time using the forgetting law, which is an important factor in the assimilation process. The above findings demonstrate that both TSKT and FSKT, which are based on assimilation and accommodation in knowledge construction, can ensure good performance in making predictions for new exercises.

5.3.2. Ablation study for each key component

We performed ablation experiments to investigate the effect of each key model component of TSKT on the prediction result. Table 4 shows the impact of two distinct (knowledge and exercises) on the KT methods in the different embedding methods. The embedding of the knowledge node, that is, “knowledge,” was chosen as the representation of the exercises, whereas “exercises” refers to the embedding of the exercise node themselves. In traditional KT methods, exercises are frequently described using knowledge concepts (Piech et al., 2015; Nakagawa et al., 2019), effectively solving the challenge of data sparsity because the number of knowledge concepts is less than the number of exercises. However, as shown in Table 4, the KT methods with exercise embedding outperforms the KT methods with knowledge embedding (“DKT + exercises” > “DKT + knowledge,” “GKT + exercises” > “GKT + knowledge,” and so on). A possible reason is that “KT + knowledge” series models ignore the difference between exercises with the same knowledge concepts (such as difficulty), resulting in a loss of substantial information regarding the exercises (Zhang et al., 2017). We also observe that “TSKT + exercise” performs better than “TSKT + knowledge”, possibly because the heterogeneous graph embedding obtains more information from the exercises (Tong et al., 2020a, 2020b), making the representation more accurate and comprehensive. Furthermore, the results of “exercises” on both the baseline and variant models outperformed “knowledge” (up to 6%), demonstrating that the exercise representations obtained by convergence of our proposed heterogeneous graphical method are effective in improving KT task performance and exhibit good versatility. Finally, we compared the learning curves of two

variant models (FSKT and TSKT) on the training and validation datasets. Fig. 5 shows that in two distinct embeddings, TSKT’s fit is more stable, gradually reduces the loss, and eventually obtains a higher AUC result, indicating that TSKT has superior fitting and generalization capacity.

We further conducted ablation experiments for the model’s three fundamental components. As shown in Table 5, there exist three variants of the TSKT model, each of which removes one key component from the complete TSKT. SKT-S is an assimilation updating model without advanced level spatial updating. TSKT-F is an accommodation updating model that excludes the forgetting feature of the temporal dimension. TSKT-H is a TSKT variant that excludes the exercise embedding in the spatial dimension. To begin with, TSKT had the best performance, demonstrating that regardless of the module that is removed (i.e., the forgetting update in temporal dimension, multilayer update in spatial dimension, or exercise heterogeneous embedding), it would degrade the method’s effect. All the three components are beneficial to KT prediction tasks. Second, TSKT-H performed the worst, indicating that the exercise heterogeneous embedding module has the most important influence on the performance of TSKT. The heterogeneous graph embedding of exercises may capture more associated information between exercises, confirming the significance of exercise embedding (Huang et al., 2021; Liu et al., 2019). Finally, the performance of TSKT-S is very similar to that of our model, which indicates that the update of adjacent nodes is able to obtain the complex relations between exercises and that the impact of higher-level nodes is small, which also verifies the result reported previously (Nakagawa et al., 2019; Tong et al., 2020a, 2020b).

5.3.3. Impact of the heterogeneous graph embedding and visualizations

To further explore the impact of the heterogeneous graph of the exercises on KT models, we reported the results of original baselines and the baselines with heterogeneous graph embeddings. For each baseline, we replaced their exercise embedding components with our heterogeneous graph embedding components. In particular, since the Assistment2015 dataset does not provide exercise indexes and exercise texts, we use knowledge concept indexes as exercise indexes and random embedding vectors as exercise texts. As shown in Table 6, the baseline with heterogeneous graph embedding (KT + H) outperforms their own original versions on AUC metrics on all datasets (“GKT + H” > “GKT,” “DKVMN + F + H” > “DKVMN + F,” “DKT + H” > “DKT,” and so on). This result demonstrates the effectiveness of our heterogeneous graph embedding method, which can be helpful for the KT task. The potential

Table 4

Impact of two embeddings (Knowledge and exercises) on the KT methods.

Embedding	Metrics	DKT	DKVMN	DKT + F	DKVMN + F	GKT	FKT	SKT	FSKT	TSKT
Knowledge	AUC	0.7682	0.7675	0.8061	0.8158	0.8017	0.8153	0.8260	0.8314	0.8393
	ACC	0.7229	0.7243	0.7570	0.7553	0.7553	0.7664	0.7716	0.7728	0.7864
Exercises	AUC	0.8132	0.8283	0.8197	0.8542	0.8307	0.8427	0.8514	0.8500	0.8617
	ACC	0.7565	0.7808	0.7599	0.7978	0.7800	0.7827	0.7935	0.7925	0.8001

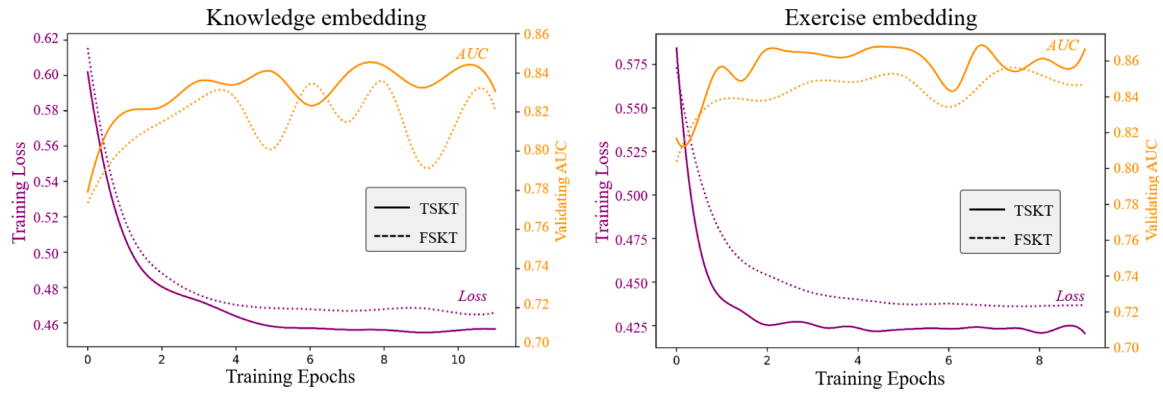


Fig. 5. Comparison of two distinct embeddings' learning efficiency on the prediction task. The purple part and orange part of the figure show the change in training loss and validating the AUC with epochs, respectively.

Table 5

Performance comparison of TSKT and its variants.

Methods	Temporal	Spatial	Heterogeneous	Knowledge embedding		Exercise embedding	
				AUC	ACC	AUC	ACC
TSKT-S	✓	–	✓	–	–	0.8614	0.8061
TSKT-F	–	✓	✓	–	–	0.8571	0.7999
TSKT-H	✓	✓	–	0.8314	0.7728	0.8500	0.7925
TSKT	✓	✓	✓	0.8393	0.7864	0.8617	0.8001

Table 6

Impact of the heterogeneous graph embedding for each baseline.

Methods	Features		Assistment2015		Eanalyst	
	Forgetting	Heterogeneous graph	AUC	ACC	AUC	ACC
DKT	–	–	0.6904	0.7275	0.8132	0.7565
DKT + H	–	✓	0.7034 (↑)	0.7289 (↑)	0.8269 (↑)	0.7555 (↓)
DKVMN	–	–	0.6969	0.7205	0.8283	0.7808
DKVMN + H	–	✓	0.6982 (↑)	0.7264 (↑)	0.8378 (↑)	0.7745 (↓)
DKT + F	✓	–	0.7027	0.7291	0.8197	0.7599
DKT + F + H	✓	✓	0.7039 (↑)	0.7317 (↑)	0.8509 (↑)	0.7864 (↑)
DKVMN + F	✓	–	0.7036	0.7246	0.8542	0.7978
DKVMN + F + H	✓	✓	0.7098 (↑)	0.7060 (↓)	0.8567 (↑)	0.7896 (↓)
GKT	–	–	0.6901	0.7269	0.8307	0.7800
GKT + H	–	✓	0.6973 (↑)	0.7263 (↓)	0.8515 (↑)	0.7825 (↑)

reasons for this are the following: the heterogeneous information of the exercises (such as text content, knowledge, and difficulty) enriches the exercises representation and acquires more information, which improves prediction accuracy (Liu et al., 2019). Furthermore, the heterogeneous graph embedding component can help the model capture the potential relationships between exercises and knowledge concepts through the constructed graph structure, which helps to avoid the data sparsity problem (Nakagawa et al., 2019).

Then, to explore the effect of heterogeneous graph embedding, we visualized the representations of knowledge and exercises using the t-Distributed Stochastic Neighbor Embedding (T-SNE) (Van Der Maaten & Hinton, 2008) method, as shown in Figs. 6 and 7. In Fig. 6, we mark the knowledge concepts of three sections with different colors for the mathematics subject, where blue, purple and yellow represent primary, middle, and high school, respectively. The knowledge concepts from the same school section are mostly concentrated in the same local area. This proves that the heterogeneous graph embedding module of the exercises in TSKT is capable of gathering similar knowledge concepts and obtaining similar representations, which aids in realizing the interplay of similar knowledge in the KT task and simulating assimilation and conformity in knowledge construction. We further explore the capacity

of the heterogeneous graph to aggregate heterogeneous information from exercises. Specifically, we chose three knowledge concepts at random and visualized representation of the exercise content via BERT encoding and the representation of the exercise nodes via heterogeneous graphs. The three different colors reflect three different concepts. The most typical approach used to enhance the representation of exercises is to encode the textual content of the exercises (Liu et al., 2019). As shown in Fig. 7 (b), comparable knowledge concepts are aggregated for the representation of the exercises after employing BERT encoding, indicating that the textual content of the exercises is an essential part of the exercises (Liu et al., 2019). However, in the comparison analysis between Fig. 7 (a), and Fig. 7(b), the exercise representation's clustering visualization significantly improves with the implementation of the heterogeneous graph method. This improvement proves that converging the heterogeneous information of the exercise can provide a more reasonable representation of the exercise (Yin et al., 2019), and the heterogeneous graph embedding method can effectively accomplish the task of exercise representation and reveal the potential relations of the exercise, improving the effect of student performance prediction.

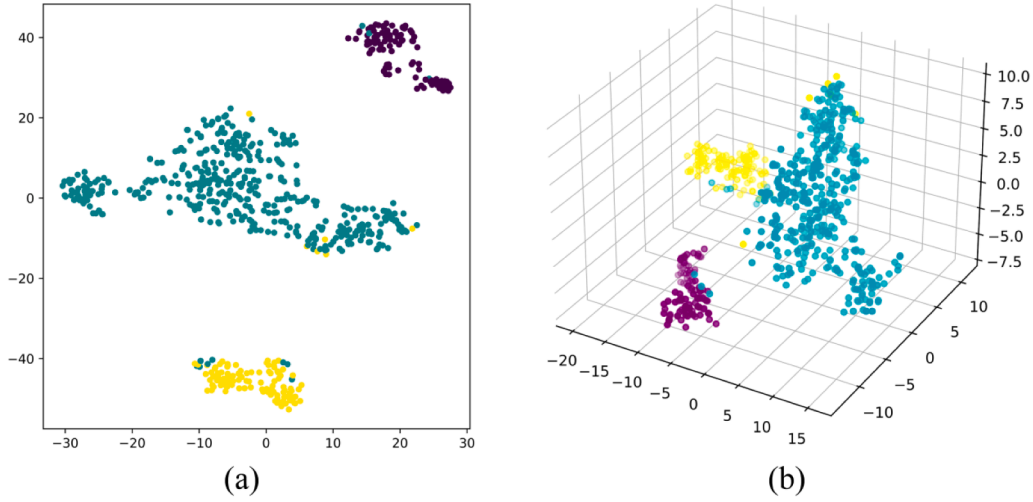


Fig. 6. Visualization of knowledge concepts in heterogeneous graph embedding in both (a) 2D and (b) 3D clustering. Different colors represent different knowledge concepts in mathematics, where blue, purple and yellow represent primary, middle, and high school, respectively. The knowledge concepts from the same school section are mostly concentrated in the same local area.

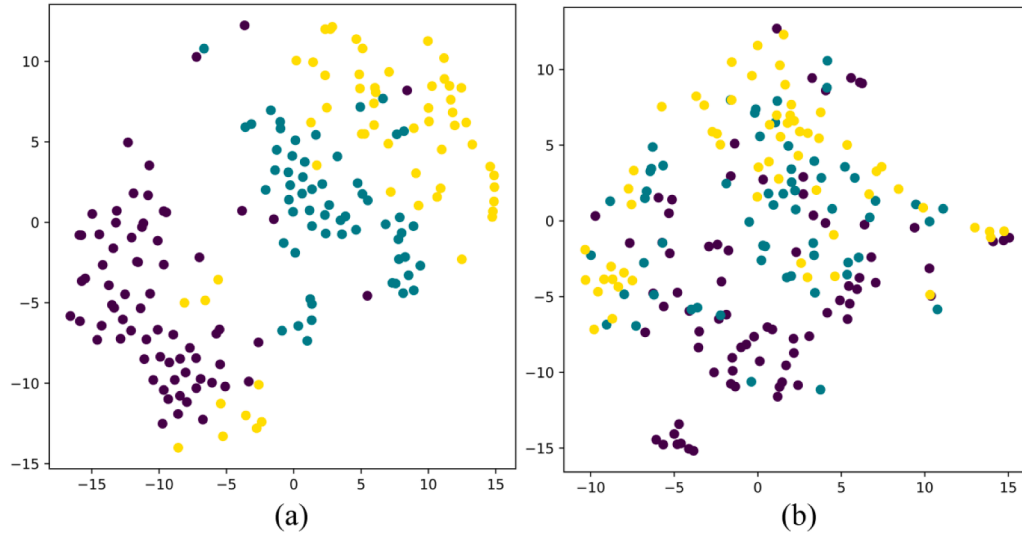


Fig. 7. Exercise visualization by using (a) heterogeneous graph embedding and (b) BERT embedding methods. The three colors represent different knowledge concepts, and the clustering of the exercise representation is better by heterogeneous graph method (a) than BERT method.

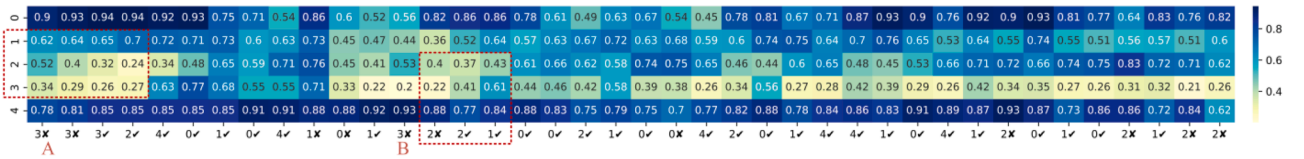


Fig. 8. Change in student knowledge state heat map in temporal dimension. Regions A and B are highlighted.

5.3.4. Spatiotemporal evolution of student knowledge state

To explore the spatiotemporal evolution of a student's knowledge state in TSKT, we visualized the interactive sequence of a random student answering 40 consecutive exercises on the Eanalist dataset. The values in the thermodynamic diagram, as shown in Fig. 8 indicate prospective knowledge mastery. The interaction sequence involves five knowledge concepts: “probability,” “central angle,” “image of quadratic function,” “function application,” and “operation of quadratic function.” When students respond incorrectly to Knowledge concept 3 (as shown in

Fig. 8 A), the color of the knowledge state deepens, showing that their knowledge mastery is lowered. Owing to the closeness between Knowledge concepts 2 and 4, when the student successfully answers the exercises relating to Knowledge concept 2, both knowledge concepts mastery will increase (as shown in Fig. 8 B). This is because the adaptation involved in knowledge construction updates the states of adjacent knowledge concepts spatially, which also verifies the mutual influence and change law of knowledge states (Nakagawa et al., 2019). Furthermore, we observed that Knowledge concept 3 demonstrated a

decreasing tendency of knowledge mastery over time owing to long-term inactivity, i.e., temporally updating the knowledge state. This result indicated that spatiotemporal dimensions had an influence on student knowledge states.

6. Conclusion

In this paper, we explore predicting future performance by tracing knowledge state evolution, guided by knowledge construction theory. Existing KT methods often overlook the influence of accommodation and assimilation on student responses, resulting in inadequate modeling of knowledge construction. To overcome this, we propose TSKT, a novel KT method that improves interpretability and performance. TSKT simultaneously traces knowledge state evolution in students' learning process along both temporal and spatial dimensions. It employs an exercise heterogeneous graph to enhance exercise representations, creating a new knowledge space with diverse exercise information and relationships. We further propose a spatial updating module that utilizes a graph neural network to update node state by current interactions and then transfers each node's influence to neighboring nodes, capturing the accommodation influence within the knowledge space. Finally, we construct a temporal updating module for updating the knowledge state through short-term memory updating and long-term memory forgetting to capture the assimilation influence of the knowledge mastery level. The spatiotemporal updating mechanism can effectively simulate the process of student knowledge state accommodation and assimilation process based on knowledge constructive theory. Extensive experiments on real-world datasets demonstrate that TSKT outperforms existing KT methods in prediction accuracy.

Although our TSKT model obtains good performance, its practical application still suffers from some limitations. The TSKT model can trace the evolution of students' knowledge status in the learning process along the temporal and spatial dimensions. However, the construction of the heterogeneous graph of exercises depends on the multi-dimensional exercise features, and the spatiotemporal updating module also needs spatiotemporal response information. Considering that we cannot capture all these features in every scenario, this leads to the limitations in the generality and applicability of the TSKT model in various scenarios or datasets. Therefore, we will further improve the general applicability and practical application of our model in future work, improving the application prospects of our TSKT model. On the one hand, we intend to leverage common features to reduce the demand for special features, thereby expanding the model's suitability across various datasets. This endeavor will facilitate the widespread adoption of our model, extending its application within a broader educational landscape. On the other hand, we will employ advanced analytical technologies, such as causal analysis (Yang et al., 2022), to identify pivotal features in the learning process while eliminating redundant or trivial attributes. Moreover, with the widespread application of smart education systems and development of artificial intelligence technology, the accompanying collection in learning process will be gradually realized (Liu et al., 2022). This development will not only promote the enrichment of collected features but also provide more possibilities for our future model research and verification. This trend contributes to the augmentation of our feature collection and offers expanded perspectives for our forthcoming model research.

Of course, there are some important directions worth exploring in the future. First, we recognize the existence of diverse relationships and structure among exercises, knowledge concepts, and students, including predecessor and sibling relations of concept-concept (Song et al., 2021), exercise-concept associations (Song et al., 2022), and multi-level relationships of student-exercise-concept (Sun et al., 2022). Further consideration of these entity relationships can enhance knowledge representation and render the spatiotemporal update module more reasonable and explicable. Second, we aim to explore the incorporation of multimodal exercise features (e.g., drawings and formulas) (Yin et al.,

2019) to capture more exercise information for comprehensive exercise representations. It is a feasible approach to explore more reasonable methods of constructing exercise heterogeneous graphs, incorporating the domain knowledge of exercises, to aggregate multimodal information and obtain more accurate exercise representations. Third, we acknowledge the significance of a spatiotemporal approach, which can be influenced by multiple factors when capturing the temporal dependencies and spatial correlations within the data. For example, depending on the interaction sequence, we can segment time series data into short-term, mid-term, and long-term subsequences, and utilize diverse techniques to extract crucial spatiotemporal features (Xu et al., 2023), such as principal component analysis or causal analysis. Lastly, the applications of KT algorithms hold great promise in revolutionizing online education systems by offering more accurate diagnostic services for assessing student knowledge status. By dynamically tracing students' cognitive levels, we can construct a learning dashboard (Aguilar et al., 2021) or design scaffolding adaptive prompts (Ueno & Miyazawa, 2017), to help students achieve high-quality personalized learning and facilitate differentiated teaching. We believe that these research directions hold significant potential for further exploration in the future.

CRedit authorship contribution statement

Huali Yang: Conceptualization, Methodology, Supervision, Writing – original draft. **Shengze Hu:** Methodology, Validation, Formal analysis, Writing – original draft. **Jing Geng:** Investigation, Resources, Writing – original draft. **Tao Huang:** Supervision, Methodology, Writing – review & editing. **Junjie Hu:** Validation, Data curation. **Hao Zhang:** Visualization, Validation. **Qiang Zhu:** Data curation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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